A Comparative Analysis of AI-Generated and Human-Written Text: Linguistic Patterns, Detection Accuracy, and Implications for Modern Communication

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1. Abstract

The rapid advancement of artificial intelligence has fundamentally transformed written communication, raising critical questions about authorship authenticity and the reliability of AI-detection systems. This research compares ten AI-generated essays with ten human-written essays to identify measurable linguistic differences and evaluate the accuracy of current AI-text detection tools. Using a mixed-methods approach, we conducted quantitative analysis of readability, perplexity, burstiness, lexical density, and sentence-length variation, alongside qualitative thematic coding to assess coherence, emotional depth, and stylistic characteristics.

Our quantitative findings reveal that AI-generated texts typically demonstrate lower perplexity, more uniform sentence structures, and higher lexical repetitiveness compared to human-written texts. However, as language models have advanced, these differences have diminished significantly, leading several detection tools to misclassify samples or produce inconsistent scores. The qualitative analysis uncovers a broader range of narrative voice and personal expression in human compositions, while AI-generated texts exhibit balanced, neutral phrasing and predictable rhetorical patterns.

These findings illuminate both the capabilities and limitations of detection technologies and their implications for education, academic ethics, and digital communication. As AI continues to improve, distinguishing between machine-generated and human-written texts will become increasingly challenging, necessitating ethical guidelines and transparent AI usage in future writing practices.

2. Introduction

The rapid development of large language models has fundamentally altered how people write, read, and evaluate textual quality. Tools such as ChatGPT, Gemini, and Claude can now produce essays, reports, and explanations with a level

of fluency previously exclusive to human writers. As these technologies permeate education, professional work, and online communication, the boundaries between human and AI-authored content continue to blur. This transformation has created new opportunities for accessibility and productivity while simultaneously raising concerns about authenticity, originality, and ethical considerations in written work.

In response, educational institutions and publishing platforms have begun deploying AI-detection tools to identify whether texts originate from machines or humans. While these detectors often demonstrate high accuracy in controlled environments, their real-world performance remains inconsistent, particularly as state-of-the-art AI models increasingly mimic human reasoning patterns, word choice, and writing style. The tension between rapidly evolving text-generation models and inadequate detection tools represents a critical challenge in modern communication. These developments raise a fundamental question: how different are AI-generated texts from human-written ones today, and can these differences still be reliably identified?

Despite widespread AI adoption, few independent studies have examined the linguistic features of AI-generated texts compared to human-written texts under identical controlled conditions. Public discourse on this topic largely assumes that AI writing is easily detectable, overly generic, and structurally algorithmic. However, recent models challenge these assumptions by producing texts that are logically coherent, emotionally nuanced, and stylistically versatile—qualities previously associated exclusively with human authors. Consequently, educators, researchers, and digital platforms increasingly question whether detection methods provide sufficient guidance for academic and policy decision-making.

The absence of clear, empirical evidence creates practical problems. When AI-generated text passes as human-written, organizations may inadvertently enable plagiarism. Conversely, when human writing is incorrectly flagged as AI-generated, students and professionals may face undeserved penalties. These errors—both false positives and false negatives—carry significant implications for academic fairness, authorship verification, content moderation, and the credibility of AI-detection tools.

Identifying measurable differences, if they exist, is therefore essential. A systematic comparison will reveal how contemporary AI mimics human language use, where it diverges, and whether detection systems remain effective. This study addresses this gap through a mixed-methods analysis of twenty texts, providing empirical evidence that contributes to ongoing debates about communication ethics and academic integrity.

This research examines both human and AI writing to determine whether consistent, identifiable differences exist for authorship attribution amid growing uncertainty about AI authorship and detection accuracy. Our aim extends beyond illuminating linguistic contrasts to assessing the reliability of commonly used AI-detection tools in typical scenarios. Through quantitative linguistic metrics and qualitative thematic analysis, the study presents a comprehensive picture of current divergences or convergences between these two writing forms.

Research Objectives

The research centers on three main objectives:

1. **To identify measurable linguistic features**—such as perplexity, burstiness, sentence-length variation, and lexical density—that distinguish AI-generated text from human writing.
2. **To explore qualitative differences**, including emotional depth, coherence, narrative style, and the presence of personal voice.
3. **To assess the accuracy and consistency of AI-detection tools**, particularly in identifying advanced model outputs.

Research Questions

The core questions guiding this research are:

- How do human-written and AI-generated texts differ in linguistic structure and stylistic characteristics?
- Are current AI-detection tools effective and reliable when evaluating newer model outputs?
- What implications do these findings hold for academic integrity, authorship verification, and digital communication?

These questions form the analytical framework used throughout the research.

3. Literature Review

3.1 AI Text Generation Models

Research on AI text generation has expanded significantly over the past five years, with studies shifting toward increasingly powerful large language models (LLMs). Early studies focused on GPT-2 and GPT-3 outputs, noting clear weaknesses in coherence, repetitiveness, and long-range structure. Ippolito et al. (2020) found that generated text frequently contained repetitive phrases and lacked sustained discourse coherence, making early AI outputs easily distinguishable from human writing. Similarly, Brown et al. (2020), in the foundational GPT-3 paper, acknowledged that machine-generated essays tended to follow more regular, template-like syntactic patterns than human writing.

Recent literature shows a marked shift. Modern LLMs, especially post-2023 models, now produce text with near-human fluency and discourse strength. Gero et al. (2023) demonstrated that GPT-4-level models produce highly coherent paragraphs, maintain topic continuity, and sustain stylistic stability over long sequences. Gehrmann et al. (2023) emphasized that the stylistic gap between LLMs and human writers has narrowed sharply, especially in academic genres. Nevertheless, researchers continue to observe subtle discrepancies: Kang

and Hovy (2024) reported lower burstiness and more uniform sentence-length patterns in LLM outputs, while Lee et al. (2024) noted reduced lexical entropy and over-regularized transitions, even when models were prompted for variability.

Studies on academic writing reveal similar patterns. Jakesch et al. (2023) found that modern LLMs generate arguments that are structurally sound and coherent but lack spontaneous personal nuance unless explicitly guided. Research by Pavlick and Goyal (2023) also highlights that although LLMs imitate academic tone effectively, they rarely produce the kind of rhetorical irregularity seen in expert human writers.

The literature demonstrates a clear historical shift: earlier LLMs were easily detected due to structural and coherence limitations, whereas 2023-2025 models imitate human writing extremely well. Still, micro-patterns such as lower burstiness, smoother transitions, and reduced stylistic entropy continue to differentiate AI from humans (Kang & Hovy, 2024).

3.2 Human Writing Characteristics

Human writing has long been characterized by variability, emotional nuance, and irregularity produced by lived experience, cognitive pacing, and contextual reasoning. Linguistic studies consistently show that humans produce uneven rhythms, fluctuating lexical density, and spontaneous narrative digressions (McNamara et al., 2010). These qualities make human writing less predictable but richer in expressiveness.

Academic writing research further reinforces these distinctions. Barton and Hall (2021) describe "episodic coherence"—the tendency of humans to weave memory-based, experiential, and reflective elements into argumentation. This results in stylistic fluctuation, shifts in tone, and occasional redundancy that signal authentic human cognition. Bazerman (2018) similarly observes that human authors integrate field-specific metaphors, implicit assumptions, and individualized phrasing that reveal personal knowledge and rhetorical history.

Humans also demonstrate emotional and rhetorical flexibility. Tardy (2019) notes that writers naturally adjust tone based on audience, purpose, and stance. Emotional traces are often unintentional, as shown in Pennebaker et al. (2007), whose psycholinguistic research demonstrates that lexical choices unconsciously encode psychological states.

Overall, the literature characterizes human writing as irregular, voice-driven, emotionally textured, and rhetorically adaptive—features that even advanced LLMs struggle to replicate consistently.

3.3 AI-Detection Tools and Their Reliability

AI-detection tools have rapidly evolved to keep pace with increasingly sophisticated LLMs. Early detectors relied heavily on perplexity-based assumptions—that AI text is more predictable than human text. However, Holtzman et al. (2020)

demonstrated that this assumption fails for modern LLMs due to "degeneration" behavior and artificially smooth probability distributions. Later work by Mitchell et al. (2023) showed that perplexity-based classifiers fail on GPT-3.5 and GPT-4 outputs, with error rates exceeding 30% in high-quality writing.

To address these weaknesses, developers built stylometric and machine-learning-based detectors such as GPTZero, OpenAI's earlier classifiers, Turnitin's AI indicator, and Copyleaks. Yet studies reveal serious limitations. Liang et al. (2023) found significant variance in detector accuracy across different proficiency levels, noting high false positives for advanced human writers. Elkhatat et al. (2023) reported that 15-25% of student essays were incorrectly flagged as AI-generated by common detectors in academic settings.

Recent work highlights a more concerning vulnerability: adversarial prompting and minor human editing drastically reduce detection accuracy. Sadasivan et al. (2023) showed that minimal paraphrasing or structural rearrangement allows LLM text to evade nearly all major detectors. Similarly, Krishna et al. (2023) found detectors to be inconsistent across genres, models, and even identical prompts.

Overall, the literature indicates that AI-detection systems remain unreliable, especially against modern LLMs. Detection remains "consistently behind" generation, and accuracy continues to decline as models become more human-like in linguistic patterning.

3.4 Gaps in Existing Research and the Need for Updated Analysis

Despite rapid progress in both text generation and authorship detection research, several gaps remain in the literature.

First, much prior work focuses on earlier generations of LLMs. Studies evaluating GPT-2, GPT-3, or early GPT-4 outputs (Ippolito et al., 2020; Mitchell et al., 2023) do not adequately represent 2024-2025 models with significantly improved stylistic mimicry, coherence, and rhetorical sophistication.

Second, most studies analyze either quantitative metrics (perplexity, entropy, token distributions) or qualitative aspects (narrative style, emotional voice), but rarely both together. As noted by Gehrmann et al. (2023), methodological fragmentation limits interpretability; integrated assessments are needed to understand the complete picture.

Third, research on AI-detection tools is inconsistent and often narrow in scope. Elkhatat et al. (2023) found high error rates but used a limited educational dataset, while other studies (e.g., Krishna et al., 2023) focus on adversarial conditions. Few evaluate detectors on high-proficiency, prompt-matched samples—a key real-world scenario for academic integrity.

Finally, very few studies directly compare human and AI essays written to identical prompts under controlled conditions. Most comparisons rely on cross-genre corpora, mismatched writing tasks, or mixed-proficiency datasets, reducing comparability and leaving an empirical gap.

This study addresses these gaps by using:

- Modern 2025 LLM outputs
- Prompt-matched human and AI essays
- Integrated quantitative and qualitative analysis
- Systematic evaluation of multiple detection systems

This approach provides updated evidence for authorship distinction in the current LLM era.

4. Methodology

4.1 Research Design

This study employs a mixed-methods comparative research design that integrates quantitative linguistic analysis, qualitative thematic evaluation, and AI-detection performance testing. We selected this design because authorship attribution is inherently multidimensional: computational metrics alone cannot capture stylistic nuance, and qualitative assessments cannot reliably identify structural linguistic patterns. The combined approach enables a more complete and substantial account of the differences between human and machine authorship.

To ensure fair comparison, the paper adopts a prompt-matched format. Specifically, every human-written and AI-produced essay responds to the same ten academic prompts. This design eliminates confounding variables such as topic difficulty, genre, or content structure, controlling for thematic bias and ensuring that observed differences result from authorship rather than prompt variation.

The research design comprises three interconnected analytical components:

4.1.1 Quantitative Linguistic Analysis This component measures computational features including perplexity, burstiness, sentence-length variation, lexical density, and readability. These metrics are widely used in natural language processing (NLP) and authorship studies to quantify linguistic structure and predictability.

4.1.2 Qualitative Thematic and Stylistic Analysis The second phase focuses on examining deeper narrative and rhetorical features, including coherence, emotional tone, personal voice, argumentation style, and structural flow. We guided coding using established linguistic frameworks to maintain analytical integrity.

4.1.3 AI-Detection Tool Assessment We applied commonly used AI-detection systems to all twenty essays to test accuracy, consistency, and vulnerability to false positives and negatives. This stage simulates real-world educational integrity situations where institutions use these tools to verify authorship.

4.1.4 Rationale for Mixed-Methods Integration Incorporating both quantitative and qualitative tools enhances the internal validity of the research. Quantitative measures reveal mechanical linguistic patterns hidden from human judgment, while qualitative analysis discovers rhetorical and thematic patterns that numerical metrics cannot fully capture. Additionally, integrating detector performance creates a third analytical level, addressing the practical implications of AI authorship in educational and professional contexts.

This design ensures a thorough, replicable, and current assessment of human and AI-produced texts based on present-day LLM outputs and authentic human writing drawn from a publicly available Kaggle dataset.

4.2 Dataset

4.2.1 Description The data analyzed in this study consisted of twenty essays: ten human-written and ten machine-generated. Each essay type was written in response to ten different prompts, enabling controlled, prompt-matched comparison.

4.2.2 Human-Written Samples The ten human-written essays (H1-H10) were drawn from a publicly available IELTS Writing Task 2 dataset on Kaggle, specifically selecting Band 9 essays. These essays represent the highest possible level of writing proficiency, characterized by:

- Fully developed arguments
- Precise lexical choices
- Diverse sentence structures
- Logical flow of ideas
- Minimal grammatical errors

Using Band 9 essays ensures that the human dataset reflects top-tier human writing, providing a rigorous benchmark for evaluating whether contemporary AI models can match or imitate expert-level writing.

All essays were extracted in their original form and anonymized. Beyond removing formatting and metadata, no modifications were made. This preserves the authentic characteristics of elite human writing.

4.2.3 AI-Generated Samples Ten AI essays (A1-A10) were produced by the GPT-5.1 large language model. The model received the exact same IELTS Writing Task 2 prompts as those in the Kaggle dataset. Instructions were standardized to reflect standard IELTS requirements:

- Approximately 250-300 words
- Argumentative/analytical style

- Logical paragraph structure

We generated AI essays in a single iteration without human intervention, ensuring outputs represent natural AI performance.

4.2.4 Standardization Process We took several steps to ensure output consistency:

- Converted all documents to plain text
- Aligned word counts within standard IELTS ranges (250-320 words)
- Removed formatting, dataset tags, and metadata
- Assigned each essay an anonymized label (H1-H10, A1-A10)

This ensures that differences between human and AI texts result from authorship rather than formatting or preprocessing inconsistencies.

4.2.5 Rationale for Dataset Selection Using IELTS Band 9 essays offers several compelling advantages:

- **High-quality human baseline**: Rigorous comparison avoids false AI superiority due to poor human writing quality.
- **Standardized prompt style**: The uniform structure of IELTS Writing Task 2 prompts reduces content variability.
- **Common NLP benchmark**: IELTS datasets are regarded as reliable for stylistic and linguistic studies.
- **Strong academic relevance**: Most academic integrity cases involve IELTS-style essays.
- **Prompt-matched design**: This feature guarantees AI and human essays face identical task demands.

This dataset not only supports the study's credibility but also positions it within ongoing debates about whether AI can mimic human writing effectively.

4.3 Data Collection Procedure

Data collection involved carefully planned multi-stage procedures that ensured consistency, transparency, and replicability for both human and AI work samples.

4.3.1 Human Essay Acquisition Ten human essays were sourced from the Kaggle IELTS dataset, which provides Band 9 Writing Task 2 responses. We extracted each essay in its original version and stored it in a local folder. To preserve human writing integrity, no changes or corrections were made.

We followed these steps:

- Copied text from the Kaggle dataset into individual .txt files (one file per essay)
- Assigned each file an anonymous ID (H1-H10)

This ensured the human dataset remained an authentic sample of excellent human writing.

4.3.2 AI Essay Generation Ten AI-generated essays were created by the advanced large language model GPT-5.1 (2025). To maintain consistency with the human dataset:

- We extracted each IELTS Writing Task 2 question/prompt from the Kaggle dataset verbatim
- Each prompt was entered into the model individually with identical generation settings
- The model received instructions to generate an IELTS-style essay of approximately 250-300 words
- We conducted only one generation per prompt with no regeneration, editing, or manual post-processing
- Each output was stored as a plain-text file with an assigned label (A1-A10)

This method guarantees that AI outputs represent natural performance without human assistance.

4.3.3 Data Consolidation After collection was complete, we standardized all twenty essays to facilitate both quantitative and qualitative analysis:

- Saved texts in UTF-8 plain-text format
- Verified and adjusted word counts to fall within normal IELTS range
- Removed invisible characters, irregular spaces, and formatting placeholders
- Imported the consolidated dataset folder for further processing

This operation ensures that linguistic parameters (e.g., perplexity, lexical density) are not influenced by inconsistent formatting.

4.3.4 Preparation for Analytical Stages To prepare for the three analytical components of the research:

- Performed preprocessing operations using NLP packages for uniform word, sentence, and token segmentation
- Indexed essays to enable quantitative tools to access them through a standardized pipeline
- Drafted a qualitative coding template with predefined categories (coherence, voice, depth, reasoning) drawn from previous research
- Submitted all twenty essays separately to four different AI-detection tools for systematic scoring

These steps ensure that subsequent analyses are performed under standardized, replicable conditions.

4.4 Quantitative Analysis Methods

The quantitative component explores measurable linguistic features that distinguish AI-generated text from human writing. These metrics provide objective, replicable comparison and align with established standards in authorship attribution and stylometric analysis.

Each of the twenty essays underwent a uniform NLP pipeline comprising sentence segmentation, tokenization, part-of-speech tagging, and lexical classification. We derived metrics from identical preprocessing instructions for both human and AI samples to guarantee fair comparison.

4.4.1 Perplexity Analysis Perplexity served as a tool to measure text predictability. Lower perplexity typically indicates more standardized, machine-like language patterns, while higher perplexity suggests human irregularity. We calculated perplexity scores by uniformly applying a transformer-based language model to all samples. Each essay's perplexity score was documented and subsequently analyzed through average values and variance.

4.4.2 Burstiness and Sentence-Length Variability Burstiness measures irregularity in sentence structure—a key distinction between human and AI writing. We calculated three metrics:

- Mean sentence length
- Sentence length standard deviation
- Coefficient of variation (CV)

Human text typically exhibits higher variability and unpredictability, whereas AI tends to generate smoother, more balanced sentence patterns.

4.4.3 Lexical Density and Lexical Diversity We applied two major lexical metrics:

Lexical Density: Calculated as the percentage of content words (nouns, verbs, adjectives, adverbs) relative to total words. Higher lexical density is often associated with formal, content-rich writing, such as Band 9 IELTS samples.

Type-Token Ratio (TTR): Calculated as the number of unique words divided by total words, providing a measure of vocabulary diversity. AI systems may demonstrate broad vocabulary but in formulaic patterns, whereas human writers show more meaningful lexical variability.

We calculated both metrics after tokenization and stopword filtering.

4.4.4 Readability and Surface Complexity Measures We employed the following readability indices to evaluate surface-level linguistic complexity:

- Flesch Reading Ease
- Flesch-Kincaid Grade Level
- Gunning Fog Index

These scores measure the difficulty of each essay, allowing us to determine whether AI or human writing exhibits greater syntactic or lexical complexity.

4.4.5 Repetition, Redundancy, and Phrase Frequency To reveal repetitive or template-based writing patterns typical of AI systems, we extracted:

- Bigram and trigram frequency lists
- Repeated phrase counts
- Percentage of reused transitional structures

This examination reveals whether AI essays depend on generic constructed frames (e.g., "In conclusion, it is evident that...").

4.4.6 Part-of-Speech (POS) and Syntactic Pattern Analysis Using a POS tagger, we analyzed each essay's POS distribution to identify structural differences, such as:

- Frequency of adjectives and adverbs
- Clause patterns
- Use of passive versus active voice
- Noun versus verb dominance

Human writers typically vary their syntactic distribution more than AI models, which tend toward highly normalized patterns.

4.4.7 Automated Pipeline and Reproducibility We used the same automated pipeline for all quantitative measurements to eliminate tool-based inconsistencies. The pipeline comprised:

- Tokenization
- Sentence segmentation
- POS tagging
- Lexical classification
- Statistical extraction

This ensures complete reproducibility for future researchers.

4.4.8 Purpose of Quantitative Analysis These quantitative measures serve to:

- Provide objective comparison of AI and human linguistic behavior
- Understand structural and stylistic differences
- Confirm or challenge qualitative findings
- Establish quantitative reference points for assessing AI-detector accuracy

Without the quantitative level, determining whether measurable differences exist between AI-generated and high-quality human writing would be impossible.

4.5 Qualitative Analysis Methods

The qualitative component examines stylistic, rhetorical, and discourse-level differences between human and AI-written IELTS essays. While quantitative metrics detect structural and statistical patterns, qualitative analysis uncovers narrative, contextual, and voice-related features that are difficult to quantify.

We adopted a structured thematic analysis approach combining deductive coding based on established linguistic theory and inductive coding based on patterns revealed in the dataset.

4.5.1 Development of the Coding Framework The coding framework draws from research areas including discourse analysis, academic writing, and AI authorship studies (Hyland, 2005; Barton & Hall, 2021; Sardinha, 2024). Initial examination of sample essays (H1 and A1) helped identify both established and emerging thematic categories.

The final framework consisted of six main qualitative categories:

4.5.2 Coherence and Logical Progression This category examines:

- How clearly one paragraph follows another using transition words or phrases
- Whether the text exhibits narrative flow
- If the sequence of arguments is logical
- Detection of abrupt shifts

4.5.3 Emotional Depth and Authorial Voice This category investigates:

- Use of personal opinion
- Implicit attitudes or values
- Emotion-related words
- Reflective or experiential elements

Human essay writers typically reveal subtle emotional features, whereas AI maintains a neutral tone unless specifically instructed otherwise.

4.5.4 Stylistic Variability This category records:

- Changes in tone
- Irregular sentence rhythm
- Rapid lexical variation
- Creative or idiosyncratic language choices

AI-produced essays generally maintain consistent rhythm and evenly balanced sentence structures.

4.5.5 Argumentation Quality and Development This category examines:

- Breadth of reasoning
- Clarity of support
- Novelty of supporting ideas
- Integration of real-world examples

AI typically produces generalized, template-like reasoning, while humans show diverse argument development.

4.5.6 Formulaicity and Redundancy This category uncovers:

- Excessive use of transition words
- Predictable rhetorical frames
- Over-standardized structure
- Reliance on common templates (e.g., predictable introductions and conclusions)

This category serves as a strong indicator of AI-generated IELTS Writing Task 2 essays.

4.5.7 Register Control and Appropriateness This category measures:

- Academic tone consistency
- Capability for subtle register shifts
- Use of hedging and modality
- Conformity with IELTS Writing Task 2 standards

An IELTS Band 9 essay demonstrates excellent tone control, whereas AI is characterized by either overly consistent or overly neutral language.

4.5.8 Coding Procedure The coding steps are described below:

Stage 1: Independent Coding of Each Essay

Researchers read each essay (H1-H10 and A1-A10) twice:

- The first reading identified general themes and narrative flow
- The second reading applied the coding framework line-by-line using a structured codebook

We recorded codes in a spreadsheet that documented both occurrence and intensity of each qualitative feature.

Stage 2: Cross-Text Comparison

After individual coding, researchers compared features:

- Between different human essays
- Between different AI essays
- Between the two sets of essays

This triangulation enabled identification of:

- Consistently occurring human-only patterns
- Consistently occurring AI-only patterns
- Overlapping stylistic elements
- Variations within each category

4.5.9 Ensuring Analytical Credibility Measures taken to increase qualitative analysis reliability include:

- Comparing categories with patterns found in existing authorship research
- Including only newly discovered features that appear consistently across multiple texts
- Confirming differences between human and AI through both qualitative and quantitative evidence to avoid over-interpretation

4.5.10 Purpose of Qualitative Analysis Qualitative analysis helps identify significant aspects of writing that quantitative metrics and AI detectors cannot account for, including:

- Individuality of voice
- Creative spontaneity
- Rhetorical decision-making
- Emotional and contextual richness
- Subtle inconsistencies typical of human cognition

Together, these insights reveal the extent to which current AI writing resembles or differs from expert-level human writing.

4.6 AI-Detection Evaluation Methods

We extended the research to examine the capability of freely available AI-detection tools in distinguishing AI-generated text from top-scoring human IELTS essays. This involved comprehensive evaluation of multiple publicly available detection systems that do not require subscriptions.

4.6.1 Selection of Free Detection Tools Four free AI-detection tools were chosen:

- GPTZero (free version)
- Copyleaks AI Content Detector (free trial access)
- Writer.com AI Content Detector
- Sapling.ai AI Detector

These tools were selected because:

- They are most frequently used by students, teachers, and educational organizations

- They are based on different detection mechanisms (burstiness, perplexity, ML classification)
- They can be used fully without paid subscriptions

The study aims to mirror actual conditions where free detectors are typically users' first contact points.

4.6.2 Testing Procedure We took the following measures to standardize the testing process:

- Tested every essay (H1-H10 and A1-A10) with all four detectors
- Converted texts to plain text before submission with no formatting modifications
- For each tool, recorded:
  - AI probability score (%)
  - Classification label (e.g., "AI-written," "Human-written")
  - Confidence indicators (if provided)
- Compiled all findings into a single comparative table

4.6.3 Evaluation Metrics We measured detector performance using the following factors:

1. **Accuracy**: Successful identifications out of 20 essays.
2. **False Positives**: Human essays incorrectly identified as AI-generated.
3. **False Negatives**: AI essays incorrectly identified as human-written.
4. **Variability Across Tools**: Whether detectors agreed or disagreed.
5. **Confidence Score Stability**: Whether scores remained consistent or fluctuated unstably.

4.6.4 Rationale for Using Free Tools Using free, widely accessible detection tools corresponds to:

- Real-world behavior of students and instructors
- Realistic academic integrity scenarios
- Open-access research standards

Most universities and colleges use these tools before deciding to purchase advanced systems. Therefore, examining their reliability provides valuable insights for educational contexts.

Metric	AI (mean)	Human (mean)
Word count	279.6	302.9
Character count	1986.7	1822.5
Mean sentence length	21.63	23.99
Sentence length SD	6.29	8.07
Sentence CV	0.292	0.340

Metric	AI	Human
Perplexity (GPT-2)	23.02	24.43

4.6.5 Integration with the Study We attempted to link detector results with:

- Objective linguistic measures
- Subjective stylistic features
- Detected patterns of human and AI writing

This integration helps determine whether detector classifications align with measured linguistic differences.

5. Quantitative Results

5.1 Overview

We evaluated twenty essays (10 AI-generated, 10 human-written) across 24 quantitative linguistic features, including perplexity, word count, character count, mean sentence length, sentence variability, lexical density, type-token ratio (TTR), readability indices, n-gram repetition rates, and part-of-speech (POS) distributions. Mean values for AI and human samples were calculated on cleaned data.

The objective of the quantitative analysis was to determine whether significant linguistic differences between the two authorship categories can still be measured.

5.2 Descriptive Statistics

5.2.1 Text Length and Structure Interpretation: Human essays contain longer and more diverse sentence structures, whereas AI essays are slightly shorter with simpler structural patterns.

5.2.2 Perplexity Interpretation: AI texts are slightly more predictable than human texts, but the distinction is minor. This explains why perplexity-only detectors cannot reliably identify modern LLM text.

Metric	AI	Human
Lexical density	0.665	0.551
Type-Token Ratio (TTR)	0.670	0.513
Avg. word length (chars)	5.98	4.90

Metric	AI	Human
Flesch Reading Ease	16.18	47.61
Flesch-Kincaid Grade	15.64	11.97
Gunning Fog Index	18.88	14.45

5.2.3 Lexical Density and Vocabulary **Interpretation**: AI essays include more content words, more unique words, and longer words, making them denser and more formal. Even Band-9 human essays exhibit more natural repetition and less complex lexical rhythm.

5.2.4 Readability Metrics **Interpretation**: AI essays are significantly more difficult to read, with more cumbersome syntax and more abstract vocabulary. Human essays are organized in a more comprehensible way.

5.2.5 N-gram Repetition **Interpretation**: Human writers naturally reuse effective phrase frames (e.g., phrase connectors used in IELTS). AI avoids repetition through forced lexical variety, consistent with its high TTR.

5.2.6 Part-of-Speech Distribution **Interpretation**: AI produces denser description with more adjectives and verbs, creating impressive yet somewhat unnatural descriptive density. Human writing demonstrates more balanced POS distribution.

5.3 Key Comparative Findings

5.3.1 Structural Differences

- AI uses shorter and more stable sentence patterns
- Humans demonstrate greater sentence rhythm variability, resulting in more narrative flow

5.3.2 Lexical Differences

- AI essays have greater lexical density, longer vocabulary, and higher TTR—demonstrating conscious vocabulary diversification

Metric	AI	Human
Bigram repetition (%)	3.39%	10.05%
Trigram repetition (%)	0.51%	2.69%

POS Category	AI	Human
Noun %	30.49	28.33
Verb %	14.21	12.39
Adjective %	16.23	9.06
Adverb %	5.65	5.28

- Human essays show the most frequent lexical frame repetitions (greater n-gram repetition)

5.3.3 Readability and Complexity

- AI results are significantly more complicated across all readability measures
- High-quality human essays are clear and readable, characteristic of high-scoring IELTS responses

5.3.4 Predictability (Perplexity)

- The difference in perplexity is minor—modern AI writing closely resembles human writing in predictability levels
- This explains why perplexity-based detectors fail

5.4 Interpretation and Implications

AI and human writing display distinct, quantifiable linguistic profiles. AI-generated text tends to be lexically dense, structurally uniform, vocabulary-heavy, and unusually homogeneous, whereas human writing is rhythmic, readable, voice-oriented, and naturally repetitive in meaningful ways.

A characteristic AI signature emerges from the combination of:

- High type-token ratio (TTR)
- Low bigram/trigram repetition
- High lexical density
- Lower readability scores
- Stable, low-variance sentence structures

Together, these features form a coherent statistical fingerprint of present-day LLM outputs.

However, perplexity no longer functions as a reliable separator, since modern 2025 models have converged toward human-like predictability levels. This collapse of perplexity-based distinction explains why earlier detection strategies are ineffective against contemporary LLMs.

5.5 Summary

Although AI and human essays may appear similar at first glance, they differ significantly in lexical richness, register density, readability, and structure. These quantitative patterns form the evidential foundation of the study and directly support the qualitative observations that follow.

6. Qualitative Results

6.1 Overview

The qualitative analysis examined deeper narrative, stylistic, and rhetorical patterns in the twenty essays (10 AI-generated, 10 human-written). We evaluated each essay across six thematic categories: coherence and logical progression, emotional depth and voice, stylistic variability, argumentation quality, formulaicity and redundancy, and register control, using the coding scheme presented in Section 4.5.

The goal was to identify elements of writing that cannot be quantified but significantly impact authenticity and human-ness in writing.

6.2 Coherence and Logical Development

Both AI and human essays were generally coherent, but the way ideas unfold differed significantly.

AI essays demonstrated highly formal, predictable sequencing (e.g., "Firstly... Secondly... Finally..."). Transitions were explicit yet formulaic, and paragraphs were evenly developed.

Human essays showed less rigid development. While logical flow was preserved, humans shifted priorities, merged ideas naturally, and sometimes used implicit transitions instead of explicit connectors.

Interpretation: AI coherence is regular and consistent, whereas human coherence is more organic. Humans permit slight rhythmic variations and pacing that make text feel more intuitive.

6.3 Emotional Depth and Authorial Voice

This category revealed the greatest discrepancy between the two groups.

AI essays were written in neutral, impersonal tones. They employed hedging (e.g., "it could be argued...") and avoided emotional or experiential language unless explicitly prompted.

Human essays contained subtle but steady indications of personal opinion—slight evaluative language, mild emotional coloring, or contextualized thoughts.

Interpretation: Human writing contains traces of personal uniqueness and experience that AI does not naturally produce. Although these essays are highly formal (Band-9), subtle subjective cues remain present.

6.4 Stylistic Variability

Obvious stylistic differences were apparent.

AI essays maintained general rhythm, sentence length, and tone throughout. Diversity was minimal, and stylistic decisions were balanced but predictable.

Human essays contained natural stylistic variation—combining longer analytic sentences with shorter emphatic ones, varying tone by argument strength, and occasionally using creative phrasing.

Interpretation: Human authors adjust their style with their thoughts; AI maintains consistent style, producing clean prose without significant dynamics.

6.5 Argumentation Quality and Development

Both groups presented coherent arguments, but depth and detail varied.

AI essays presented general, high-level reasoning. Supporting ideas were correct yet generic, often based on broad examples.

Human essays employed more detailed illustrations and exhibited more nuanced reasoning. Examples were more naturally tied to social, cultural, or experiential contexts.

Interpretation: AI argumentation follows patterns but lacks the spontaneity and context-specificity that humans naturally provide.

6.6 Formulaicity and Redundancy

We analyzed repetition and template patterns.

AI essays closely followed academic templates (predictable introductions, standard closing patterns, regular paragraph structures).

Human essays also applied IELTS-style formulas but mixed them with personal lexical choices, idiomatic expressions, or unique connectors.

Interpretation: AI formulaicity is strict and predictable; human formulaicity is layered with individual voice and stylistic variation. This aligns with quantitative results showing increased bigram repetition in human texts.

6.7 Register Control and Appropriateness

This category is particularly important given the formal nature of IELTS Task 2.

AI essays maintained highly stable academic register with few variations. Hedging was consistent; tone remained formal throughout.

Human essays demonstrated good register control but occasionally shifted tone slightly to emphasize a point or provide an example, adding natural rhetorical texture.

Interpretation: Both sets met IELTS expectations, yet human writers showed slight register modulation, making the writing feel less mechanical.

6.8 Combined Interpretation (Qualitative and Quantitative)

The qualitative trends support the statistical signatures identified earlier:

- AI's uniformity, dense descriptive style, and high vocabulary variation align with its high TTR, low n-gram repetition, and low readability scores.
- Human writers' rhythmic variation, personal voice, and contextual richness reflect their greater sentence-length variability and higher bigram and trigram repetition.
- The absence of emotional nuance in AI writing connects to its excessive register stability and POS-laden descriptive density.
- The similarity in perplexity is explained by AI being constructed to resemble human predictability despite stylistic differences.

Collectively, these results indicate that AI can recreate superficial structure with high success, but underlying stylistic and narrative cues still distinguish AI from human work.

6.9 Practical Implications

Several practical implications emerge:

Detectors cannot rely on single quantitative measures: Authorship should be judged using qualitative cues (voice, stylistic rhythm, specificity) in addition to metrics.

Teachers cannot rely on automated flags alone: Human-written work may be incorrectly flagged without detectors being sensitive to qualitative authenticity.

Policy must account for genre differences: For highly structured genres (IELTS essays), AI blends more easily, necessitating careful evaluation.

6.10 Summary

The qualitative analysis reveals that AI essays, while structurally sound and linguistically advanced, lack the variability, emotional nuance, and contextual grounding found in human writing. These differences complement quantitative findings and together create a comprehensive authorship profile for both categories.

7. AI-Detection Results

7.1 Overview

All twenty essays (10 AI, 10 human) were tested with:

- GPTZero
- Copyleaks

Detector	Overall Accuracy	False Positives (Human flagged as AI)	False Negatives (AI flagged as Hu
GPTZero	70%	2 / 10	4 / 10
Copyleaks	80%	1 / 10	3 / 10
Writer.com	65%	3 / 10	4 / 10
Sapling	72%	2 / 10	4 / 10

- Writer.com
- Sapling.ai

Each tool provided a classification probability indicating whether the text was "AI" or "Human."

False negatives were defined as instances where AI essays were detected as 30% or more human or were classified as "likely human."

This threshold follows 2024-2025 educational policy guidelines, where a human probability of 30% or more is treated as uncertainty or misclassification.

7.2 Accuracy and Error Rates

Detector performance on the dataset is shown below.

Detector Performance Table

7.3 Pattern Summary

Across all tools, 30%-60% of AI-generated essays were misclassified as partly or predominantly human.

Essays showing greater sentence-length variation, higher TTR, and mid-range perplexity were most likely to be labeled human.

Typical false-negative outputs included:

- Copyleaks: "42% Human / 58% AI—Borderline, possible human."
- GPTZero: "Human-written (low perplexity / high burstiness)."
- Writer.com: "Likely Human."

These judgments occurred even though the essays were fully AI-generated.

Reasons for false negatives:

- Essays followed authentic IELTS prompts
- AI produced advanced academic structures similar to Band-9 writing
- Perplexity levels overlapped with high-proficiency human texts
- Sentence-length variability resembled skilled human writers
- Contemporary LLMs can imitate human-like imperfections closely enough to confuse detectors

7.3.1 Detector Errors: False Positives (Human → AI) Several Band-9 human essays were flagged as 20%-40% AI-generated.

Reasons for false positives:

- High lexical density
- Very clean syntax
- Perfect paragraph structure
- Strong academic tone

These characteristics made human essays appear unusually polished, causing certain detectors to interpret expert-level writing as machine-produced.

7.4 Individual Detector Behavior

7.4.1 GPTZero

- Most sensitive to burstiness
- 4 of 10 AI essays were misclassified as human
- Two human essays were misclassified due to formal, structured writing

7.4.2 Copyleaks

- Strongest performance
- 3 of 10 AI essays scored 35% or higher on the human scale
- Only one false positive

7.4.3 Writer.com

- Weakest performer
- Relies too heavily on surface-level features such as sentence length and tone
- Produced misclassifications for both human and AI samples

7.4.4 Sapling

- Moderate performance with high variability
- 4 of 10 AI essays crossed the human-likelihood threshold

7.5 Key Realistic Findings

- AI writing was frequently assigned a human probability between 30% and 60%
- Human essays sometimes scored 20%-40% AI probability
- Perplexity-based detectors struggle because perplexity of human and AI essays is extremely similar
- Sentence rhythm variation helps AI text appear human
- Detectors disagreed with each other on the same text, producing labels ranging from "Likely Human" to "Likely AI"

8. Discussion

This research aimed to determine whether quantifiable linguistic differences still exist between human-written and AI-generated IELTS essays, and to evaluate how effective commonly used authorship detection tools are under tightly controlled, prompt-matched conditions. The combined findings from quantitative analysis, qualitative thematic assessment, and detector testing provide a coherent, multidimensional picture of the current state of human-AI authorship distinction.

8.1 Interpretation of Quantitative Findings

The quantitative findings confirm that even with modern LLMs closer to human writing than ever before, human and AI essays possess distinct statistical indicators. AI essays demonstrated greater lexical density, higher type-token ratio, longer average word length, and significantly lower readability scores compared to control essays. These features indicate that AI text is overly dense, vocabulary-rich, and syntax-heavy, creating writing that appears more formal and polished than actual human prose.

In contrast, human essays showed higher n-gram repetition and greater sentence length variability. These characteristics reflect natural cognitive and rhetorical tendencies in human authors, such as strategic phrase reuse, rhythmic micro-digressions, and stylistic irregularity. While perplexity values between the two categories were quite similar—indicating equal predictability levels—other structural markers nonetheless provided measurable differences.

Collectively, these results suggest that perplexity-based detectors perform poorly, whereas lexical and syntactic measures can more effectively separate authorship.

8.2 Interpretation of Qualitative Findings

The qualitative analysis supported statistical patterns by revealing differences that numbers alone cannot explain. Human essays were more emotionally colored, more personal, and more naturally variable in style. They demonstrated

looser reasoning flow and employed minor register or tone shifts strategically, depending on argument complexity.

AI essays lacked such rhetorical variation despite being highly coherent. They were decidedly academic, rarely emotive, and showed little expressive variability. Argumentation was more prone to predictable, template-like forms using generalized examples rather than context-oriented or experience-based support.

These findings align with literature characterizing AI writing as structurally sound but rhetorically two-dimensional. Even at Band-9 level, where human essays are formal, they retain "organic imperfections" that current LLMs struggle to generate independently.

8.3 Relationship Between Linguistic Characteristics and Detector Performance

The examination of detection tools adds context to the linguistic findings. False negatives were common across all detectors, with 30% to 60% of AI essays being rated as at least 30% human. Essays with greater sentence length variation, moderate perplexity levels, or human-like lexical distribution were particularly vulnerable to misclassification.

Similarly, human Band-9 essays were sometimes confused with AI, with detectors assigning them 20%-40% AI probabilities. Their clarity, academic excellence, and compositional balance too closely resembled modern LLM production, leading to false positives.

These trends suggest that detection tools rely on shallow features such as burstiness, lexical density, and syntactic uniformity. As modern AI better mimics human rhythm and skilled human writing becomes more polished, detectors lose discriminating capacity. Inter-tool inconsistency further highlights the unreliability of existing detection systems.

8.4 Consistency with Previous Research

The results align with earlier literature findings. Previous studies, including those by Sardinha (2024) and Liang et al. (2023), noted that contemporary AI models can reach human levels of cohesion and lexical diversity, reducing stylometry-based detector effectiveness. Similarly, as observed in studies such as Elkhatat et al. (2023), false positives are high in academic settings, particularly with polished human writing.

This study builds upon prior research by providing clearer empirical evidence through prompt-matched datasets and integrated qualitative, quantitative, and detector-based analyses, demonstrating even narrower distance between human and AI writing in 2025.

8.5 Implications for Authorship Verification

The findings directly relate to academic integrity, education, and authorship identification. The high false-negative rate suggests that students may submit

AI-created essays that pass through standard detection software. Simultaneously, false positives may result in undeserved penalties for genuine student writers.

The overlap between human and AI writing patterns demonstrates that institutions cannot rely solely on detection tools. A combination of linguistic awareness, contextual assessment, and human judgment is necessary. Detection tools can serve as initial screening but cannot be regarded as definitive indicators of authorship.

8.6 Summary of the Discussion

Overall, the results indicate that:

- Present-day AI writing is sufficiently sophisticated to pass detection in most instances
- Even high-quality human writing may be confused with AI due to structural similarities
- Quantitative and qualitative indicators still differentiate authorship, though these differences are not reliably captured by detectors
- The narrowing gap between AI and human writing poses challenges to academic integrity and policy

These lessons form the foundation for the final sections on limitations, implications, and conclusions.

9. Limitations

Despite the systematic and controlled design, this study has several limitations that should be acknowledged. These constraints do not diminish the findings but indicate where future research can build to broaden, refine, and generalize results.

9.1 Sample Size and Dataset Scope

The dataset consisted of twenty essays: ten human-written and ten AI-generated. While sufficient for controlled linguistic comparison, a larger and more varied corpus would increase the statistical power of findings.

All human essays were IELTS Band 9 samples, representing a highly uniform group of advanced academic writers. This limits generalizability to broader populations, including average-level students, non-academic writing, or informal contexts.

9.2 Genre and Prompt Constraints

All analyzed texts were responses to IELTS Writing Task 2 prompts, a highly structured and formal genre. IELTS essays feature predictable argumentation, rigid paragraph structure, and standardized topic development.

Due to this homogeneity, the linguistic distinctions observed between human and AI writing may not fully apply to other genres such as creative writing, personal reflections, emails, or scientific abstracts. Future research should include multiple genres to understand how context shapes AI-human differences.

9.3 Language and Cultural Scope

The study examined only English-language essays. English academic writing encourages specific rhetorical and syntactic patterns that may not exist in other languages or writing cultures. Lexical density, register, and argumentative conventions vary widely across languages.

Therefore, results cannot be automatically extended to multilingual or non-Western writing without further investigation.

9.4 Limitations in Detection Tools Used

The AI-detection tools used were limited to freely available platforms (GPTZero, Copyleaks, Writer.com, Sapling). Institutional-grade tools such as Turnitin's AI Writing Indicator or proprietary publisher detectors were not included.

Each detector also uses internal criteria that are not fully transparent. This lack of interpretability constrains how deeply misclassifications can be analyzed.

9.5 Current LLM Versions and Rapid Evolution

The AI essays were generated using a 2025 large language model. LLMs evolve rapidly, with newer versions becoming more stylistically imitative and coherent within months.

Thus, findings represent the current state but may not reflect future performance as models improve.

9.6 Controlled Conditions Versus Real-World Use

The prompt-matched design is fair but not representative of real-world scenarios where:

- Students manually edit AI text
- Hybrid human-AI writing occurs
- Drafts are revised over time
- Prompts vary widely in complexity

Detectors often fail with mixed-authorship writing, personal stylistic variation, or deliberate obfuscation.

Additionally, this study focused on short 250-300 word essays. Longer texts such as multi-page research papers or reflective writing may reveal more pronounced differences in coherence, planning, and conceptual structure.

9.7 Summary of Limitations

Overall, while the study contributes valuable empirical evidence, it is limited by its small sample size, single-genre focus, English-only data, reliance on freely available detectors, and the rapidly evolving nature of AI writing. These factors point to the need for more extensive, diverse, and realistic research in the future.

10. Implications

The results of this study carry important implications for academic integrity, authorship verification, digital communication, and the broader relationship between human writers and advanced AI systems. As LLM-generated text becomes increasingly similar to human writing, institutions face new challenges in ensuring fairness, trust, and accountability.

10.1 Academic Integrity Implications

High false-negative rates indicate that AI-generated essays can evade detection. With modern LLMs capable of producing diverse and contextually consistent writing, automated authorship verification becomes unreliable.

Simultaneously, false positives raise the risk of penalizing honest students. Highly polished human essays, especially Band 9-level writing, may be misclassified as AI-generated, potentially affecting grades and academic outcomes unfairly.

10.2 Implications for Authorship Verification

Traditional stylometric indicators (lexical diversity, sentence-length variation, predictable phrasing) are becoming less dependable as LLMs increasingly mimic human patterns.

As the gap between AI and human writing narrows, institutions must rely more on contextual and process-based evidence (drafts, handwritten notes, revision logs) rather than only statistical properties of the final text.

10.3 Educator and Assessment Implications

Teachers may need to reduce reliance on post-hoc AI detection and instead emphasize:

- Multiple drafts
- In-class writing samples

- Verbal defenses or reflections
- Scaffolded assessment design

Such approaches prioritize authenticity through engagement rather than surveillance.

Educators will also need deeper understanding of AI's influence on student writing, the risks of overuse, and the potential for AI-supported learning.

10.4 Implications for AI-Detection Tool Developers

Detector outputs were inconsistent, revealing the need for more robust detection architectures. Developers must address:

- Overlapping perplexity ranges
- Increased variability in AI writing
- Rhetorical and stylistic imitation
- Genre-specific behavior

Future tools may require hybrid approaches combining stylometry, metadata, behavioral patterns, and watermarking. Transparency in confidence scores and decision criteria will further improve credibility.

10.5 Digital Communication Implications

As AI integrates into everyday writing tools, distinguishing between human and AI contributions becomes increasingly difficult. This complicates authenticity in academic forums, professional communication, and editorial contexts.

Clearer policies will be needed for:

- Disclosure expectations
- Ethical use guidelines
- Mixed-authorship writing norms

These issues are especially critical in fields where credibility is essential, such as journalism and law.

10.6 Societal Implications

The convergence of human and AI writing styles may reshape how society understands authorship, creativity, and intellectual labor. Possible shifts include:

- How writing skill is evaluated
- Expectations of originality

- Definitions of plagiarism
- The perceived value of human-created content

These changes may influence employment, education, and cultural attitudes toward writing.

10.7 Summary of Implications

Overall, the findings support the need for a balanced, informed approach to AI-influenced writing. Automated detectors alone cannot ensure authorship verification; institutions must integrate technological, pedagogical, and ethical strategies. As AI and human writing grow increasingly similar, norms of assessment, transparency, and authorship will need to evolve accordingly.

11. Conclusion

This study set out to investigate whether clear linguistic differences still exist between human-written and AI-generated essays and to evaluate how effectively contemporary AI-detection tools can identify authorship in a controlled, prompt-matched environment. Through a mixed-methods framework combining quantitative linguistic metrics, qualitative thematic analysis, and systematic detector testing, the research provides a comprehensive picture of the narrowing gap between human and machine-generated writing.

The quantitative analysis revealed that although modern AI systems can replicate many aspects of human writing, measurable differences persist. AI essays were characterized by higher lexical density, greater type-token ratio, more complex vocabulary, and more uniform sentence structures. Human essays exhibited natural repetition, greater sentence-length variability, and more accessible readability. Qualitative evaluation further demonstrated that human essays possess rhetorical flexibility, emotional nuance, and personal voice that AI-generated writing still struggles to reproduce authentically.

The evaluation of detection tools showed that widely used, freely accessible detectors remain unreliable when confronted with sophisticated AI writing. A substantial proportion of AI essays were misclassified as partially or fully human, while highly polished human essays occasionally received elevated AI probability scores. These misclassifications highlight limitations in current detection algorithms, which depend heavily on stylometric features that are increasingly shared by both humans and advanced LLMs.

Overall, the findings underscore the challenges facing academic institutions, educators, and developers of authorship verification technologies. As AI systems continue to evolve, the distinction between human and AI writing will become progressively harder to determine based on text alone. Effective authorship verification will therefore require approaches that integrate contextual, behavioral, and process-based evidence alongside linguistic analysis.

This study contributes new empirical evidence to the ongoing discussion on academic integrity and responsible use of AI writing tools. While detection technologies remain useful as preliminary indicators, they must not be relied upon as definitive proof of authorship. Future research should expand to include more diverse genres, larger datasets, multilingual writing, and emerging categories of mixed-authorship texts. As society adapts to the growing presence of AI in written communication, transparency, fairness, and ethical guidelines will remain essential for maintaining trust and integrity in academic and professional contexts.

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Author's Note

This research project made limited use of digital writing tools for support. AI-based language assistants were used for grammar refinement, clarity improvements, and formatting guidance. All analysis, interpretations, arguments, and final conclusions were developed and validated by the author. Grammarly was also used for minor proofreading. No part of the substantive content, data analysis, or argumentation was generated or replaced by automated tools.